

10 (**short**) A particle of mass m slides on a frictionless horizontal table. The particle is attached by a long light inextensible string, which passes through a small frictionless hole, to a second mass, also of mass m , attached to a spring of stiffness k , as shown in Fig. 7.

(a) The particle is moving in a circular orbit of radius R such that the tension in the spring is zero. Determine the tension in the string. [2]

(b) A tangential impulse is applied to the orbiting particle such that it moves out to a maximum radius of $2R$.

(i) Calculate the velocity of the particle at this radius. [5]

(ii) Calculate the new tension in the string. [3]

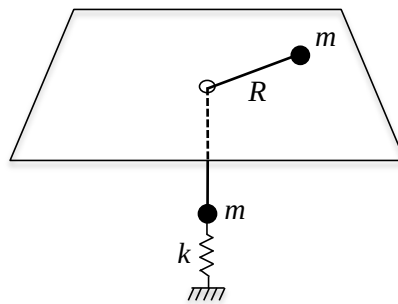


Fig. 7